



Working Draft 1
MEF W139.1

**Internet Access Product Schema and
Developer Guide**

July 2025

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is subject to change.**

MEF W139.1

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1 List of Contributing Members

The following members of the MEF participated in the development of this document and have requested to be included in this list.

Editor Note 1: This list will be finalized before Letter Ballot. Any member that comments in at least one CfC is eligible to be included by opting in before the Letter Ballot is initiated. Note it is the MEF member that is listed here (typically a company or organization), not their individual representatives.

2 Abstract

This MEF Standard implements business requirements defined by MEF W164 - Internet Access Product Attributes and Use Cases [18]. It consists of this schema guide and its associated software artifacts (JSON Schemas) that define and describe the product-specific information used in LSO Cantata and LSO Sonata APIs for a set of Business Functions - specifically, Product Offering Qualification, Quote, Product Ordering, and Product Inventory – for the Internet Access product.

The document starts with an overview of LSO Cantata, LSO Sonata, and Internet Access services. It then provides a basic information model for the MEF Internet Access Attributes. The final sections describe the Data Model described by JSON Schemas associated with this specification.

This Standard normatively incorporates the following files by reference as if they were part of this document, from the GitHub repository <https://github.com/MEF-GIT/MEF-LSO-Sonata-SDK>, commit id: TODO

productSchema/ip/

- TODO

Also included in the GitHub repository is a Postman file that contains informative examples illustrating use of the Internet Access. This file is not part of this standard but is referred to in Appendix A.

- documentation/productSchema/ip/internetAccess/MEF 139.1 - Appendix A.postman_collection.json

3 Terminology and Abbreviations

This document does not define any new terms or definitions. All of the terms defined in the standards below are included in this document by reference:

- MEF 164 - Internet Access Product Attributes and Use Cases [18]
- MEF 55.1 Lifecycle Service Orchestration (LSO): Reference Architecture and Framework [4]
- MEF 55.1.1 Amendment to MEF 55.1: Reference Architecture and Framework - Terminology [5]
- MEF 57.2 Product Order Management Requirements and Use Cases [6]
- MEF 61.1 IP Service Attributes [7]
- MEF 79.1 - Product Offering Qualification Management Business Requirements and Use Cases [9]
- Mplify 150 Installation Place and Service Site Management - Business Requirements and Use Cases [17]

4 Compliance Levels

The key words "**MUST**", "**MUST NOT**", "**REQUIRED**", "**SHALL**", "**SHALL NOT**", "**SHOULD**", "**SHOULD NOT**", "**RECOMMENDED**", "**NOT RECOMMENDED**", "**MAY**", and "**OPTIONAL**" in this document are to be interpreted as described in BCP 14 (RFC 2119 [2], RFC 8174 [3]) when, and only when, they appear in all capitals, as shown here. All key words must be in bold text.

Items that are **REQUIRED** (contain the words **MUST** or **MUST NOT**) are labeled as [Rx] for required. Items that are **RECOMMENDED** (contain the words **SHOULD** or **SHOULD NOT**) are labeled as [Dx] for desirable. Items that are **OPTIONAL** (contain the words **MAY** or **OPTIONAL**) are labeled as [Ox] for optional.

5 Numerical Prefix Conventions

This document uses the prefix notation to indicate multiplier values as shown in Table 1.

Decimal		Binary	
Symbol	Value	Symbol	Value
k	10 ³	Ki	2 ¹⁰
M	10 ⁶	Mi	2 ²⁰
G	10 ⁹	Gi	2 ³⁰
T	10 ¹²	Ti	2 ⁴⁰
P	10 ¹⁵	Pi	2 ⁵⁰
E	10 ¹⁸	Ei	2 ⁶⁰
Z	10 ²¹	Zi	2 ⁷⁰
Y	10 ²⁴	Yi	2 ⁸⁰

Table 1 Numerical Prefix Conventions

6 Introduction

This chapter start with an overview of LSO Sonata and LSO Cantata. Then the document contains the following sections:

- An overview of the Internet Access Product (section 7)
- Data Model Design Principles (section 8)
- Order Milestones (section 9)
- The relationships between the Internet Access Product and other entities (section 10)
- Details of the Data Model (section 11)

6.1 Overview of LSO Cantata and LSO Sonata

MEF 55.1 [4] is the Reference Architecture for Lifecycle Service Orchestration (LSO) of MEF-defined services. It defines seven LSO Interface Reference Points (see Figure 1) that are abstract interconnection points between different entities, either within the Service Provider domain (intra-domain) or between the Service Provider and other business entities (inter-domain). One of these LSO Reference Points is LSO Cantata, which defines the abstract interconnection point between a Subscriber (Buyer) and a Service Provider (Seller), and another is LSO Sonata, which defines the abstract interconnection point between a Service Provider (Buyer) and an Operator (Seller). It is at these Interface Reference Points - LSO Cantata and LSO Sonata - that the Buyer and the Seller interact to orchestrate business transactions for the different Business Functions. Inter-provider Business Functions include Address Qualification, Site Query, Product Offering Qualification, Quote, Product Ordering, Product Inventory, Trouble Ticketing, and Billing. In the context of this document, the following 4 business functions are relevant as ones exchanging product information:

- Product Offering Qualification, MEF 79.1 [9]
- Quote, MEF 80.1 [10]
- Product Ordering, MEF 57.2 [6]
- Product Inventory, MEF 81 [11]

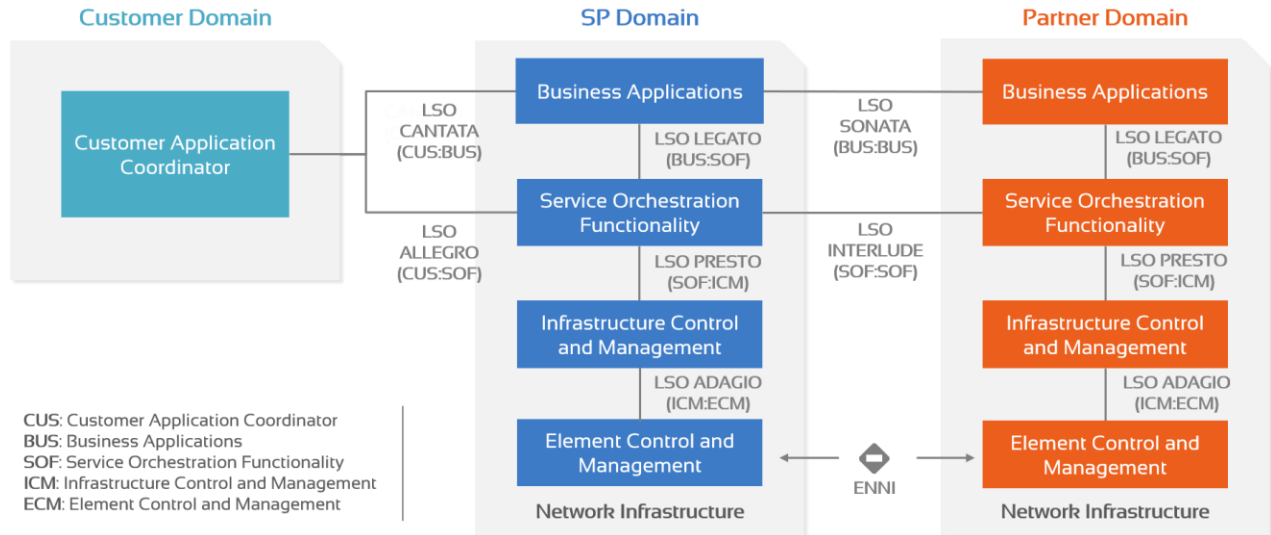


Figure 1 LSO Cantata and LSO Sonata Reference Diagram

The mutual access to Business Functionalities is automated via APIs at the LSO Cantata and LSO Sonata Interface Reference Points, which are standardized by MEF as LSO Cantata and LSO Sonata APIs, and which are made available by MEF in a series of releases of the LSO Cantata SDK and LSO Sonata SDK. The APIs are standardized by following the API and Developer Guide documents:

- Product Offering Qualification, MEF 87.1 [12]
- Quote, MEF 115.1 [13]
- Product Ordering, MEF 123.1 [16]
- Product Inventory, MEF 116.1 [14]

The LSO Cantata and LSO Sonata APIs comprise two parts: a product-agnostic API and a set of product-specific data models, as shown in Figure 2.

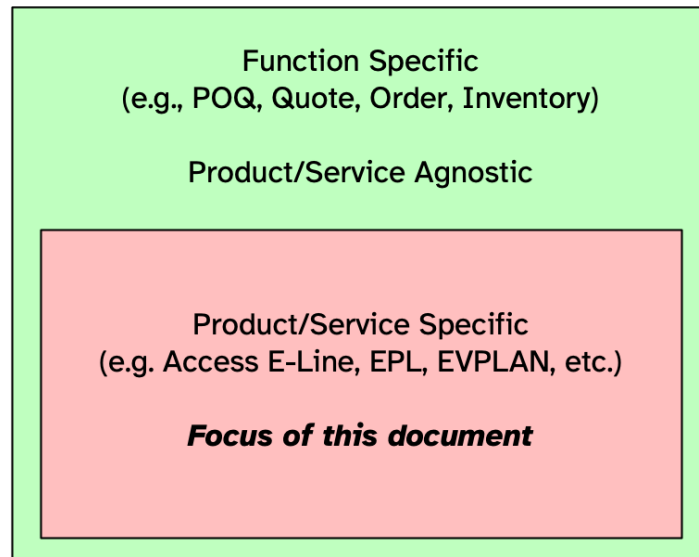


Figure 2 LSO Cantata and LSO Sonata API Structure

This document describes the product-specific Data Model for MEF Internet Access.

7 Overview of Internet Access

An Internet Access is a Subscriber IP service which means it is provided to an end-user (the Subscriber) by a Service Provider. A Subscriber can be an enterprise, a mobile operator, an IT system integrator, a government department, etc. An Internet Access provides the Subscriber with connectivity to the global Internet. In this case, the Service Provider is acting as an Internet Service Provider.

This section defines a set of generic Internet Access Product Offerings (Use Cases) that can be used by Sellers as the basis for their Product Offerings. Each of Product Attributes described in Section 11 indicates which of these Use Cases the Product Attribute is applicable to.

7.1 MEF Broadband Internet Access, Basic

This Product is the simplest type of Internet Access Product and is commonly used for Residential Internet Access. Internet Access is provided via a Broadband Network which can be, for example, DOCSIS, PON, Satellite, Cellular. The Product includes the Internet Access Device necessary for the type of Broadband Network. Most basic Internet Access Products provide options for bandwidth, but some also provide additional options such as fixed IP addresses, cellular backup, various security options, etc.

7.2 MEF Broadband Internet Access, Advanced

This Product is similar to the Basic Internet Access Product. It is delivered over a Broadband Network and includes the Internet Access Device, but it is often more business-focused and designed for small businesses with a single location or a small number of locations with relatively modest technical requirements. It usually provides some options not available in Basic Internet Access such as multiple Public IP Addresses, and better Service Level Agreements, better availability, etc.

7.3 MEF Dedicated Internet Access, no Router

This Product is a Dedicated Internet Access Product usually delivered on a wavelength (or part of a wavelength) on a fiber optic interface although it could be delivered over other media such as microwave, satellite, etc. It is designed for larger organizations that have substantial technical requirements for their Internet Access including (for example) substantial flexibility in deployment of IP addresses, advertising routes, dual homing, redundancy, etc. For this product, the Buyer provides the router.

7.4 MEF Dedicated Internet Access, with Router

This Product is the same as the “MEF Dedicated Internet Access, no Router” but the Seller is providing the router. The router can be physical or virtual. Therefore, the required and optional Product Attributes are the same as the previous Use Case with the addition of Product Attributes related to the router and features that can be enabled on the router.

8 Data Model Design Principles and Assumptions

An attribute for a Product can have a value that is a simple datatype such as an integer or string (or list of simple datatypes) or a value that is an object with multiple properties such as a LocalNetworkAddresses or a composition of objects such as Bandwidth. Within this document, each simple value (integer, string, boolean, etc.) is referred to as a Product-Specific Attribute.

The data model for the Internet Access product configuration is expressed as a JSON schema based on JSON schema draft 7 [1] and encoded in YAML. This schema accompanies this document. It can also be found in the SDK release in the following path:

schema\productSchema\ip\internetAccess\internetAccess.yaml

In the schema the following naming conventions are used:

- class and type names are UpperCamelCase,
- Product Attribute and property names are lowerCamelCase,
- enumeration values are defined using UPPER_SNAKE_CASE.

The data model is defined with variations, depending on:

- Business Function - a given Product-Specific Attribute may, for example, be classified as Optional for the Create POQ request; while it may be classified as Mandatory for the Create Product Order request.
- Applicable Use Cases: It indicates with which of the Use Cases described in section 7 the Product Attribute is appropriate for use.

To avoid multiplication of schemas resulting from possible combinations, there is only one schema provided that fulfills all the requirements possible, leaving the rest to be implemented and checked by business logic, and not by the API contract.

On top of the standard-defined model, the Buyer and the Seller may agree on which attributes are required, predefined, or used in a given context. This can be achieved, for example with the use of the Product Catalog API.

Note: A CHANGE request cannot change a single Service Attribute. The Buyer must send a full product configuration, including all Mandatory Attributes and all Optional Attributes that were previously specified by the Buyer (in an INSTALL request or previous CHANGE request). Any Optional Service Attributes that are not specified in a CHANGE request are reset to their default value.

In the figures, some classes' attributes or further class structures are skipped for diagram readability. This is denoted by the "<<skipped>>" clause.

For readability of the diagrams, the default multiplicity of relations is 0..1.

Some requirements define Product Attributes as mutually exclusive. This means that either one or the other must be provided, but not two (or more) of them at the same time. This is defined in the schema using the “oneOf” statement in the “required” section of the type definition.

For example, *BgpExtendedCommunity* has 4 attributes (among others): *regularType*, *regularValue*, and *extendedType*, *extendedValue*, but only one pair of them must be set at the same time. This part of the schema that defines this requirement looks as follows:

```
oneOf:  
  - required: [regularType, regularValue]  
  - required: [extendedType, extendedValue]
```

In the following sections, where applicable, this information is provided after the table with the attributes.

Tables listing the attributes have the following columns:

- Name - attribute name as present in the schema file,
- Type - the data type of the attribute. All additional constraints are also listed in this column if they are defined in the schema (ex. minItems, maxItems, minimum, etc.)
 - List attributes are designated by square bracket “[]” next to the type name. E.g. “Ipv4OrIpv6Prefix[]” means the attribute is a list of objects of type “Ipv4OrIpv6Prefix”
- M/O - specifies if the attribute is mandatory or optional to provide.
- Description - description of the attribute.

9 Order Milestones

The Service Provider (Seller) can provide Product-Specific Product Order Item Milestone notifications to the Buyer on the status of a Product Order as a sequence of Milestones for that Order as they are achieved. For ordering Internet Access, the following milestones are commonly used (a Service Provider may support some or all of these milestones, and not all milestones are applicable for all orders).

The Milestone Name in the first column is included in ProductOrderItem.milestone and ProductSpecificProductOrderItemMilestoneEventPayload.milestoneName in the Product Order API (see MEF 123.1 [16]).

Note: Milestones and their notifications are independent of the Product Order Item's state.

Milestone Name	Description
SITE_SURVEY_SCHEDULED	Site Survey Scheduled
SITE_SURVEY_COMPLETE	Site Survey Complete
PLANNING_COMPLETE	Planning Complete
FIRM_DELIVERY_DATE_PROVIDED	Firm Delivery Date Provided
AWAITING_MUNICIPAL_APPROVAL	Awaiting Municipal Approval
MUNICIPAL_APPROVAL_GRANTED	Municipal Approval Granted
AWAITING_LANDLORD_APPROVAL	Awaiting Landlord Approval
LANDLORD_APPROVAL_GRANTED	Landlord Approval Granted
CONSTRUCTION_STARTED	Construction Started
CONSTRUCTION_COMPLETED	Construction Completed
AWAITING_ACCESS	Awaiting Site Access Permission (for the end-to-end test)
ACCESS_DENIED	Site Access Denied (for the end-to-end test).
AWAITING_WIRING	Awaiting Installation of Inside Wiring by Landlord
WIRING_COMPLETE	Installation of Inside Wiring by Landlord Complete
EQUIPMENT_DISPATCHED	Equipment Dispatched
EQUIPMENT_DELIVERED	Equipment Delivered
EQUIPMENT_INSTALLED	Equipment Installed
E2E_TESTING_SCHEDULED	End-to-End Testing Scheduled
E2E_TESTING_COMPLETED	End-to-End Testing Completed
E2E_TESTING_FAILED	End-to-End Testing Failed.

Table 2 Order Milestones for Internet Access

10 Relationships Between Entities

This section describes the relationships (and their constraints) between the products and between the products and places.

Internet Access is a location-specific product. The installation location can be provided, depending on the use case, with:

- Place Relationship – for Broadband Basic and Broadband Advanced use cases, OR
- Product Relationship to Service Access Interface – for Dedicated no Router and Dedicated with Router use cases.

[R1] The Buyer **MUST** specify the Installation Place either with Place Relationship or Product Relationship for POQ, Quote, and Order.

[R2] For Broadband Internet Access Basic and Advanced, the Buyer **MUST** use the Place Relationship to specify the Installation Place.

[R3] For Dedicated Internet Access with and without Router, the Buyer **MUST** use the Product Relationship to a Service Access Interface to specify the Installation Place.

Product	Place Relationship Role	INSTALL	CHANGE
Internet Access	INSTALL_LOCATION	Mandatory	Mandatory

Table 3 Place Relationship

[R4] The Role field (role) of the Related Place (RelatedPlaceRefOrQueryWithSubUnit or RelatedPlaceRefOrQuery) type **MUST** contain the INSTALL_LOCATION value shown in the Place Relationship Role column in Table 3.

Source Product	Relationship Type	INSTALL	CHANGE	Target Product	Multiple Allowed at POQ and Quote?
Internet Access	CONNECTS_TO_SAI	Mandatory	Mandatory	Service Access Interface	No

Table 4 Product Relationship

11 Internet Access Attributes

This section provides a guide to the detailed model of the Internet Access product in all flavors. MEF W164 defines sets of requirements on Internet Access Attributes. Some of them are fulfilled by already existing API functions and parameters (marked as “Envelope” in Table 5), and some of them are part of the Internet Access product schema (“Payload”), defined by this standard.

When an attribute is covered by API features’ respective requirements defined by the API standard are applicable. These requirements are not mentioned in this document, as they are part of a broader business flow defined in these APIs and should not be taken out of the context.

Attribute	ENV/Payload	Broadband				Dedicated			
		POQ	Quote	Order	Inventory	POQ	Quote	Order	Inventory
Installation Location	Envelope	M	M	M	M	N/A	N/A	N/A	N/A
	Envelope	N/A	N/A	N/A	N/A	M	M	M	M
Demarcation Point	Envelope	N/A	N/A	O	M	N/A	N/A	N/A	N/A
Buyer Contact Information	Envelope	O	O	O	O	O	O	O	O
Requested Lead Time	Envelope	O	O	O	O	O	O	O	O
Requested Installation Date	Envelope	O	O	O	O	O	O	O	O
Contract Term	Envelope	O	O	O	O	O	O	O	O
Delivery Mechanism	Payload	M	M	M	M	M	M	M	M
Service Access Interface	Payload	O	O	M	M	N/A	N/A	N/A	N/A
VLAN	Payload	N/A	N/A	N/A	N/A	O	O	M	M
List of Features	Payload	O	O	O	O	O	O	O	O
Bandwidth	Payload	O	O	O	O	O	O	O	O
Guaranteed Bandwidth	Payload	O	O	O	O	O	O	O	O
Billing Model	Payload	O	M	M	M	O	M	M	M
Local Network Addresses	Payload	O	O	O	O	O	O	O	O
Provider Independent IP Addresses	Payload	O	O	O	O	O	O	O	O
Autonomous System Number	Payload	O	O	O	O	O	O	O	O
IP Address Type	Payload	O	M	M	M	O	M	M	M
Public Addresses Prefix Length	Payload	O	O	O	O	O	O	O	O
Public Address Block	Payload	O	O	O	O	O	O	O	O
Internet Access Transit Addresses	Payload	N/A	N/A	N/A	N/A	O	O	O	O

Table 5 Internet Access All Attributes Summary

Table 5 lists all Internet Access Product attributes and summarizes realization by either Envelope or Payload and requirements or applicability per Use case and API, as defined by MEF 164.

[O1] The Buyer **MAY** include a Product Attribute in a request unless an explicit requirement indicates otherwise.

[R5] All Product Attributes specified in a Product Order **MUST** be included in the Product Inventory.

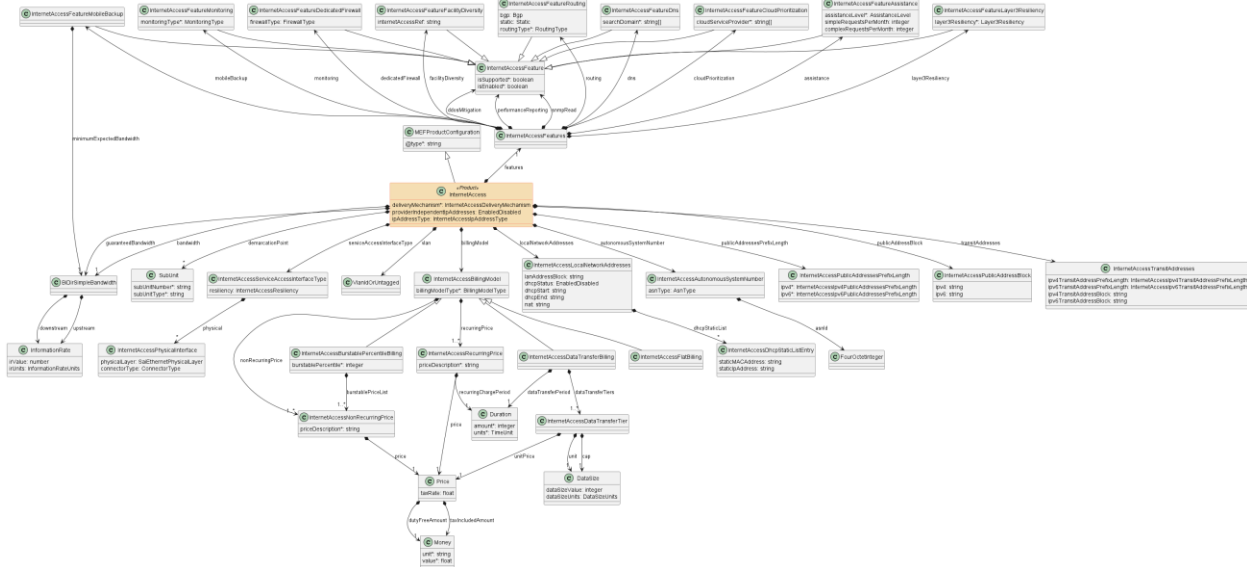


Figure 3 Internet Access model Overview

Figure 3 presents an overview of the Internet Access Model. It has the Routing Protocols attribute collapsed to avoid complexity.

Table 6 presents the list of Internet Access object attributes. The following chapters list all attributes and provide details about their implementation and model. The business-level information on the attributes is provided in the respective chapters of MEF 164.

File: schema\productSchema\ip\internetAccess\internetAccess.yaml

Name	Type	M/O
demarcationPoint	SubUnit[]	O
deliveryMechanism	InternetAccessDeliveryMechanism	M
serviceAccessInterfaceType	InternetAccessServiceAccessInterfaceType	O
vlan	VlanIdOrUntagged	O
features	InternetAccessFeatures	M
bandwidth	BiDirSimpleBandwidth	M
guaranteedBandwidth	BiDirSimpleBandwidth	O
localNetworkAddresses	InternetAccessLocalNetworkAddresses	O
providerIndependentIpAddresses	EnabledDisabled	O
autonomousSystemNumber	InternetAccessAutonomousSystemNumber	O
ipAddressType	InternetAccessIpAddressType	O
publicAddressesPrefixLength	InternetAccessPublicAddressesPrefixLength	O
publicAddressBlock	InternetAccessPublicAddressBlock	O
transitAddresses	InternetAccessTransitAddresses	O

Table 6 Internet Access

11.1 Installation Location

This attribute is covered by the LSO API Envelope and is covered in section 10.

11.2 Demarcation Point

Applicable Use Cases: Broadband Internet Access Basic, and Advanced

This Product Attribute presents the demarcation point between the responsibility of the Buyer and the responsibility of the Seller upon delivery of the Internet Access Product.

Demarcation Point is modeled as a list of SubUnits.

File: schema\common\common.yaml

Name	Type	M/O	Description
subUnitNumber	string	M	The discriminator used for the subunit, often just a simple number but may also be a range.
subUnitType	string	M	The type of subunit e.g. BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF.

Table 7 SubUnit

[R6] For Dedicated Internet Access Products, the *demarcationPoint* **MUST NOT** be provided.

[R7] For Broadband Internet Access Products, the *demarcationPoint* **MUST** be included in the Product Inventory.

[O2] For Broadband Internet Access Products, the *demarcationPoint* **MAY** be specified in the Product Order.

Example:

```
{
  "demarcationPoint": [
    {
      "subUnitType": "FLOOR",
      "subUnitNumber": "3"
    },
    {
      "subUnitType": "ROOM",
      "subUnitNumber": "305"
    },
    {
      "subUnitType": "RACK",
      "subUnitNumber": "7"
    },
    {
      "subUnitType": "CHASSIS",
      "subUnitNumber": "2"
    }
  ]
}
```

```
435     "subUnitType": "SLOT",
436     "subUnitNumber": "5"
437   },
438   {
439     "subUnitType": "PORT",
440     "subUnitNumber": "12"
441   }
442 ]
443 }
```

11.3 Buyer Contact Information Product

Applicable Use Cases: All

This Product Attribute specifies a list of contacts where each element in the list indicates the contact role and the contact information for an individual or group responsible for a specified function, action, or role. A list of contact roles is found in Table 38 in MEF 57.2 [6].

A Buyer Administrative Contact is typically the representative of a single administrative entity (such as a university, a business enterprise or a business division) responsible for managing the service contract and handling administrative tasks related to the service such as service changes, billing, and payments, and, when necessary, management and registration of the Buyer's IP Addresses.

This attribute is covered by the use of:

- *ProductOfferingQualificationItem.relatedContactInformation* in POQ [12],
- *QuoteItem.relatedContactInformation* in Quote [13],
- *ProductOrderItem.relatedContactInformation* in Product Order [16],
- *Product.relatedContactInformation* in Product Inventory [14]

The *RelatedContactInformation* Type defines the *role* attribute that specifies the contact role in the given context. The additional *buyerAdministrativeContact* role must be supported.

[R8] The Buyer and the Seller **MUST** support the additional *buyerAdministrativeContact* role of *RelatedContactInformation*.

Example:

```
464 "relatedContactInformation": [
465   {
466     "number": "12-345-6789",
467     "emailAddress": "john.example@buyer.example.com",
468     "role": "quoteItemTechnicalContact",
469     "organization": "Buyer Co.",
470     "name": "John Example",
471     "numberExtension": "1234"
472   }
```

],

11.4 Requested Lead Time

Applicable Use Cases: All

This Product Attribute is used by the Buyer to indicate the desired lead time for installation of the Internet Access Product.

This attribute is covered by the use of:

- *ProductOfferingQualificationItem.installationInterval* in POQ
- *MEFQuoteItem_Common.requestedQuoteItemInstallationInterval* in Quote
- *QuoteItem.quoteItemInstallationInterval* in Quote

The *installationInterval* is provided by the Seller in the POQ response. The Buyer requests a *requestedQuoteItemInstallationInterval* in the Quote, and the Seller responds with the *quoteItemInstallationInterval* that the Seller can meet, which may be shorter or longer than the *requestedQuoteItemInstallationInterval*.

Example:

```
"installationInterval": {  
  "amount": 10,  
  "units": "businessDays"  
},
```

11.5 Requested Installation Date

Applicable Use Cases: All

This Product Attribute specifies the installation date requested by the Buyer.

This attribute is covered by the use of:

- *MEFProductOrderItem_Common.requestedCompletionDate* in Product Order

Example:

```
"requestedCompletionDate": "2025-05-25T21:32:28.826Z",
```

11.6 Contract Term

Applicable Use Cases: All

This Product Attribute specifies the terms of the contract for the Internet Access Product.

This attribute is covered by the use of:

- *MEFQuoteItem_Common.requestedQuoteItemTerm* in Quote
- *MEFQuoteItem_Common.quoteItemTerm* in Quote

- *MEFProductOrderItem_Common.requestedItemTerm* in Product Order
- *MEFProductOrderItem_Common.itemTerm* in Product Order

The Buyer requests a *requestedQuoteItemTerm* in the Quote and the Seller responds with the closest *quoteItemTerm* that the Seller can meet, which may be different than the *requestedQuoteItemTerm*.

Example:

```
"requestedQuoteItemTerm": {
  "duration": {
    "amount": 12,
    "units": "calendarMonths"
  },
  "endOfTermAction": "autoRenew",
  "name": "Yearly Subscription"
},
```

11.7 Delivery Mechanism

Applicable Use Cases: All

The value of this Product Attribute describes the type of equipment provided by the Seller as part of the Internet Access Product.

File: schema\productSchema\ip\internetAccess\internetAccess.yaml

Type: InternetAccessDeliveryMechanism

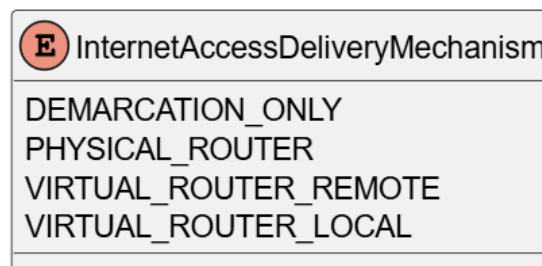


Figure 4 InternetAccessDeliveryMechanism

The Delivery Mechanism values are defined as follows:

- *DEMARCATIION_ONLY* – The product is delivered to the customer without IP routing. It is the Buyer's responsibility to provide the mandatory routing function.
- *PHYSICAL_ROUTER* – The product delivered to the Buyer includes a physical IP Router installed at the customer's premises.
- *VIRTUAL_ROUTER_REMOTE* – The product delivered to the Buyer includes an IP routing function, which is executed in a virtual environment, not at the customer premises. The network connectivity from the Buyer's location to the virtual router provides transport only.

- *VIRTUAL_ROUTER_LOCAL* – The product delivered to the Buyer includes an IP routing function running in a virtual environment at the Buyer's premises. The virtual environment (frequently referred to as a universal CPE, white-box CPE, or edge device) is provided by the Seller and can be pre-existing or can be delivered with the Internet Access Product.

[R9] The Internet Access Delivery Mechanism Product Attribute **MUST** be specified for POQ, Quote, and Order.

11.8 Service Access Interface

Applicable Use Cases: Broadband Internet Access, Basic and Advanced

When the Internet Access Product is not delivered at an existing Service Access Interface (see section 10), the value of this Product Attribute describes important aspects of the Service Access Interface to be delivered by the Seller.

File: schema\productSchema\ip\internetAccess\internetAccess.yaml

Type: InternetAccessServiceAccessInterfaceType

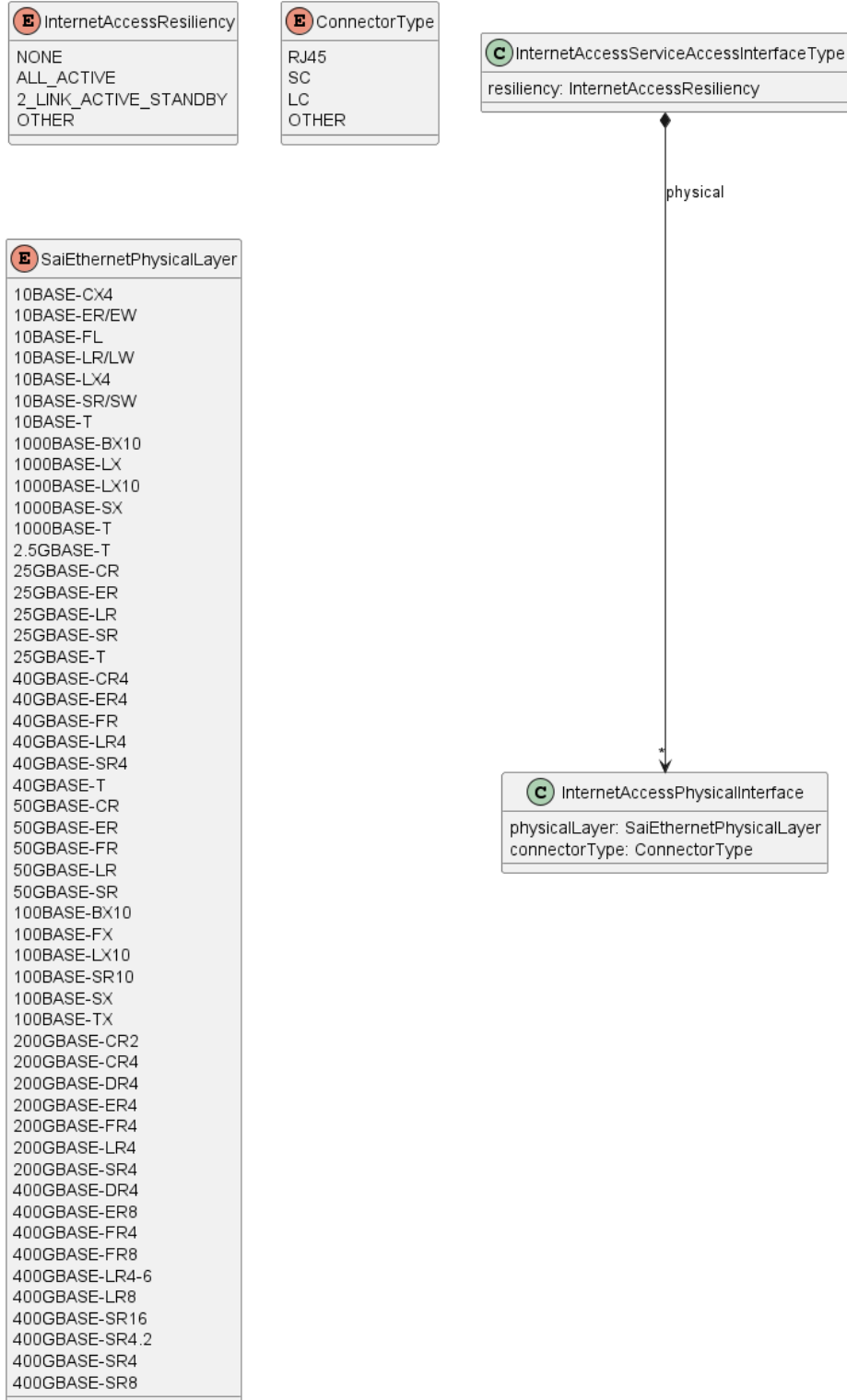


Figure 5 Service Access Interface Attribute

Name	Type	M/O	Description
physical	InternetAccessPhysicalInterface[]	O	Describes the physical aspects of the Service Access Interface delivered by the Seller
resiliency	InternetAccessResiliency	O	Describes the configuration for resiliency at the Service Access Interface

Table 8 InternetAccessServiceAccessInterfaceType

Name	Type	M/O	Description
physicalLayer	SaiEthernetPhysicalLayer	O	Specifies a physical layer
connectorType	ConnectorType	O	Indicates the connector type,

Table 9 InternetAccessPhysicalInterface

[R10] For Broadband Internet Access Basic and Advanced, the value of the *serviceAccessInterfaceType* **MUST** be included in the Product Order.

[O3] For Broadband Internet Access Basic and Advanced, the value of the *serviceAccessInterfaceType* **MAY** be included in the POQ or Quote.

[R11] For Dedicated Internet Access with and without Router, the value of the *serviceAccessInterfaceType* **MUST NOT** be included in any function.

The *Resiliency* values are defined as follows:

- NONE* - No resiliency
- ALL_ACTIVE* - indicates that Link Aggregation is used, and all the specified physical interfaces can be active and carrying traffic simultaneously.
- 2_LINK_ACTIVE_STANDBY* - indicates that Link Aggregation is used and that there are two physical interfaces and only one of them can be carrying traffic.
- OTHER* - indicates that the Buyer and Seller have agreed on a resiliency approach other than Link Aggregation.

[R12] The value *NONE* **MUST** be selected if and only if the list of physical interfaces (*physical*) contains exactly one entry.

[R13] If the value *2_LINK_ACTIVE_STANDBY* is, then the list of physical interfaces (*physical*) **MUST** contain exactly 2 entries.

[R14] If the list of physical interfaces (*physical*) contains more than two entries, then the option for resiliency **MUST** be either *ALL_ACTIVE* or *OTHER*.

Example:

```
{
  "physical": [
    {
      "physicalLayer": "1000BASE-T",
```

```
        "connectorType": "RJ45"
    }
],
"resiliency": "NONE"
}
```

11.9 VLAN

Applicable Use Cases: Dedicated Internet Access, with and without router

The value of this Product Attribute is the C-tag VLAN ID that is used for the Internet Access at the Service Access Interface. It is an integer with a value between 1 and 4094. The value 0 indicates that untagged Ethernet frames are used for Internet Access.

File: schema\common\ip\ipCommon.yaml

Type: VlanIdOrUntagged

[D1] The *vlan* attribute **SHOULD** be omitted for POQ and Quote.

[R15] The *vlan* attribute **MUST** be specified for Product Order.

Requirement [R15] indicates that the Buyer is required to tell the Seller in the Product Order whether it is using untagged frames (value 0) or VLAN-tagged frames (value greater than 0).

11.10 Features

Applicable Use Cases:

- Broadband Internet Access, Advanced
- Dedicated Internet Access, with and without router

The value of this Product Attribute indicates the major functional and protocol features provided for the Internet Access.



Figure 6 Internet Access Features

File: schema\productSchema\ip\internetAccess\internetAccess.yaml

Type: InternetAccessFeatures

Name	Type	M/O
assistance	InternetAccessFeatureAssistance	O
cloudPrioritization	InternetAccessFeatureCloudPrioritization	O
ddosMitigation	InternetAccessFeature	O
dedicatedFirewall	InternetAccessFeatureDedicatedFirewall	O
dns	InternetAccessFeatureDns	O
facilityDiversity	InternetAccessFeatureFacilityDiversity	O
layer3Resiliency	InternetAccessFeatureLayer3Resiliency	O
mobileBackup	InternetAccessFeatureMobileBackup	O
monitoring	InternetAccessFeatureMonitoring	O
performanceReporting	InternetAccessFeature	O
routing	InternetAccessFeatureRouting	O
snmpRead	InternetAccessFeature	O

Table 10 InternetAccessFeatures

Each defined feature is modelled as a separate named attribute of *InternetAccessFeature* type or its extension, when feature specific configuration is required. *InternetAccessFeature* allows for basic configuration of a feature:

Name	Type	M/O	Description
isSupported	boolean	O	Indicates whether support of the feature by the Seller is required.
isEnabled	boolean	O	Indicates whether Buyer requests that the Feature is enabled.

Table 11 InternetAccessFeature

[R16] For each feature, if the value of *isEnabled* is *True*, then the value of *isSupported* **MUST** be *True*.

A Buyer can request that a particular feature be supported but not enabled (at the time of the request). If the Buyer requests that a particular feature be enabled, support for the feature is clearly needed.

If a feature is not specified, it is equivalent to specifying the feature with *isSupported* = *False* and *isEnabled* = *False*.

Example:

```

"features": {
  "assistance": {
    "isSupported": true,
    "isEnabled": true,
    "assistanceLevel": "CO_MANAGED",
    "simpleRequestsPerMonth": 5,
    "complexRequestsPerMonth": 1
  },
  "cloudPrioritization": {

```

```
627     "isSupported": true,
628     "isEnabled": false,
629     "cloudServiceProvider": []
630 },
631 "ddosMitigation": {
632     "isSupported": false,
633     "isEnabled": false
634 },
635 "dedicatedFirewall": {
636     "isSupported": true,
637     "isEnabled": true,
638     "firewallType": "BASIC"
639 },
640 "dns": {
641     "isSupported": true,
642     "isEnabled": true,
643     "searchDomain": [
644         "mef.example.com",
645         "example.com"
646     ]
647 },
648 "facilityDiversity": {
649     "isSupported": true,
650     "isEnabled": true,
651     "internetAccessRef": "internetAccess-0001",
652 },
653 "layer3Resiliency": {
654     "isSupported": true,
655     "isEnabled": true,
656     "layer3Resiliency": "WIRELESS_BACKUP"
657 },
658 "mobileBackup": {
659     "isSupported": true,
660     "isEnabled": true,
661     "minimumExpectedBandwidth": {
662         "downstream": {
663             "irValue": 100,
664             "irUnits": "MBPS"
665         },
666         "upstream": {
667             "irValue": 20,
668             "irUnits": "MBPS"
669         }
670     }
671 },
672 "monitoring": {
```

```

    "isSupported": true,
    "isEnabled": true,
    "monitoringType": "PROACTIVE"
  },
  "snmpRead": {
    "isSupported": false,
    "isEnabled": false
  }
}

```

11.10.1 Assistance

Definition: The level of Installation, Operation, and Configuration Changes done by Seller

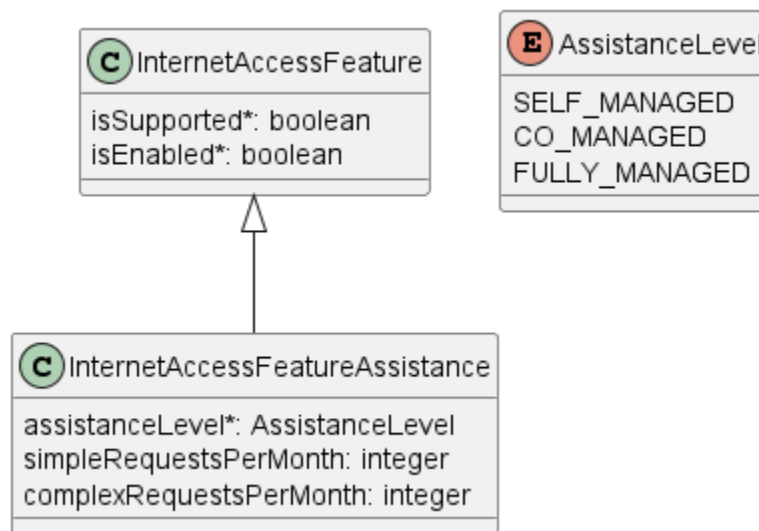


Figure 7 Assistance Feature

Feature Parameters for POQ:

- *assistanceLevel*

Feature Parameters for Quote:

- *assistanceLevel*
- If *assistanceLevel* is *CO_MANAGED* or *FULLY_MANAGED*
 - simpleRequestsPerMonth*
 - complexRequestsPerMonth*

Feature Parameters for Order: Same as Quote

11.10.2 Cloud Prioritization

Definition: The Seller enables prioritization, traffic congestion control, and SLAs for cloud traffic.

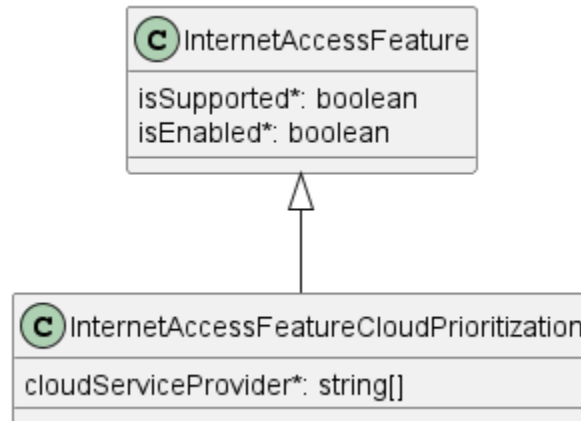


Figure 8 Cloud Prioritization

Feature Parameters for POQ:

- cloudServiceProvider

Feature Parameters for Quote: Same as POQ

Feature Parameters for Order: Same as POQ

11.10.3 DDOS-Mitigation

Definition: The Seller provides enhanced protection of the customer network from Distributed Denial of Service (DDoS) attacks.

Feature Parameters for POQ: No additional parameters

Feature Parameters for Quote: No additional parameters

Feature Parameters for Order: No additional parameters

11.10.4 Dedicated Firewall

Definition: The Seller provides a firewall service implemented on a dedicated device.

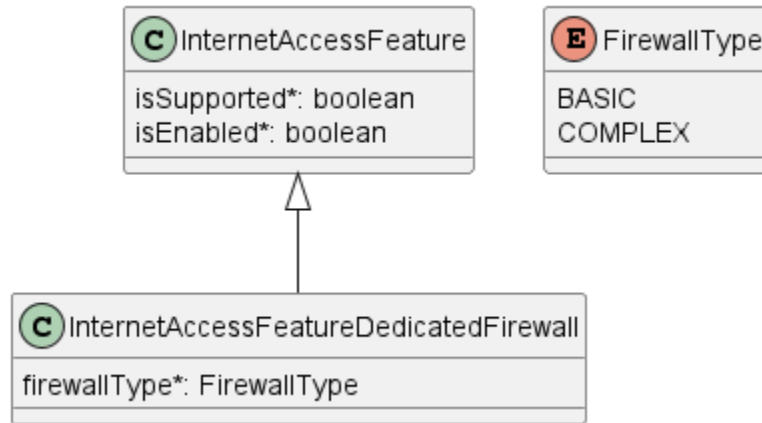


Figure 9 Dedicated Firewall Feature

Feature Parameters for POQ:

- firewallType

Feature Parameters for Quote: Same as POQ

Feature Parameters for Order: Same as POQ

The value Basic denotes a simple Firewall based on IP 5-tuples and/or Access Control Lists. The value Complex denotes more complex Firewall capabilities such as Deep Packet Inspection.

11.10.5 DNS

Definition: The Seller is providing a Domain Name Service to translate Domain Names to IP addresses.

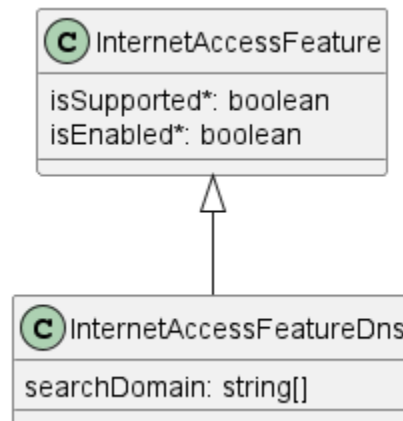


Figure 10 DNS Feature

Feature Parameters for POQ: No additional parameters

Feature Parameters for Quote: No additional parameters

Feature Parameters for Order:

- searchDomain* – a possibly empty ordered list of domain names, specified as strings, that are sequentially appended to the specified domain name if the specified domain name does not include a dot (.), until the first successful DNS lookup.

11.10.6 Facility Diversity

Definition: The Seller ensures that this Internet Access Product has no common points of failure with another Internet Access Product.

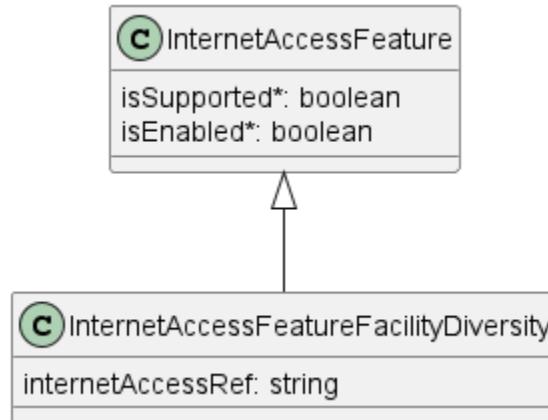


Figure 11 Facility Diversity Feature

Feature Parameters for POQ: No additional parameters

Feature Parameters for Quote: No additional parameters

Feature Parameters for Order:

- internetAccessRef* - Identifier for the other Internet Access Product

11.10.7 Layer 3 Resiliency

Definition: The Seller provides Layer 3 resiliency for the access connection to the Buyer.

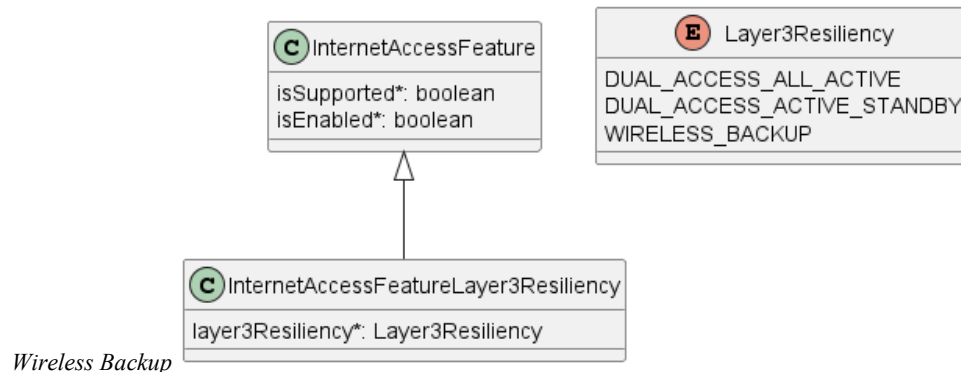


Figure 12 Layer 3 Resiliency Feature

Feature Parameters for POQ:

- layer3Resiliency

Feature Parameters for Quote: Same as POQ

Feature Parameters for Order: Same as POQ

[R17] If *layer3Resiliency* feature is configured with *isSupported=true* and *layer3Resiliency=WIRELESS_BACKUP*, then the *mobileBackup* feature **MUST** also have the *isSupported=true*

[R18] If *layer3Resiliency* feature is configured with *isEnabled=true* and *layer3Resiliency=WIRELESS_BACKUP*, then the *mobileBackup* feature **MUST** also have the *isEnabled=true*

11.10.8 Mobile Backup

Definition: The Seller provides a method for backing up the Internet Access via a 4G or 5G cellular service.

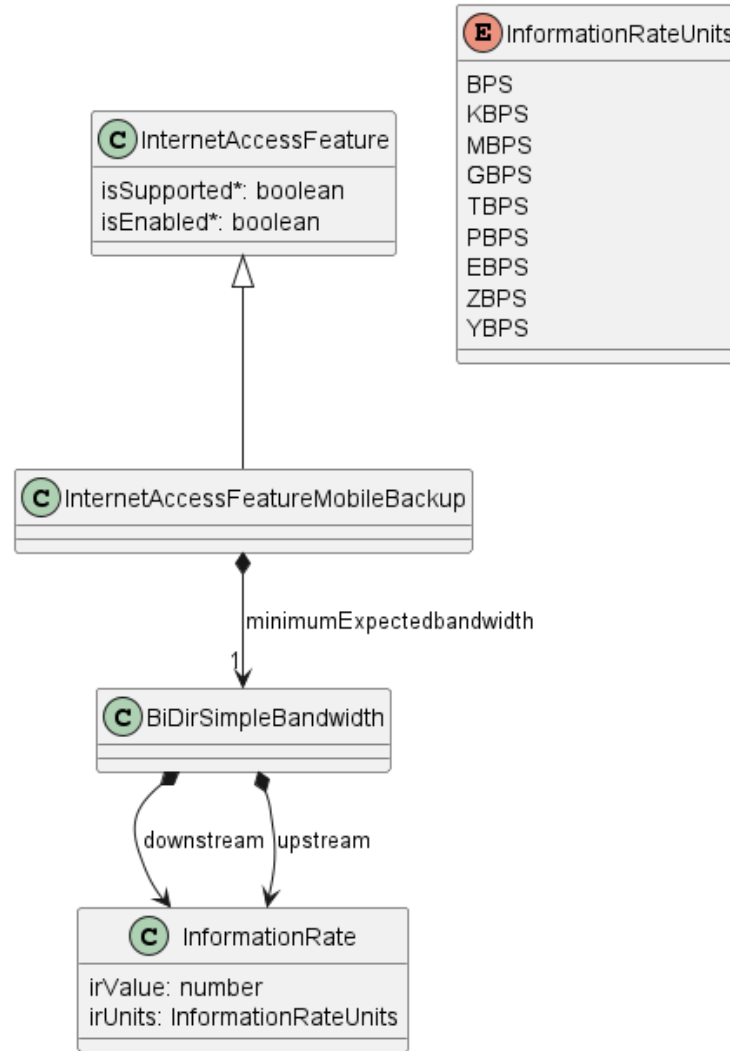


Figure 13 Mobile Backup Feature

Feature Parameters for POQ:

- *minimumExpectedBandwidth*

Feature Parameters for Quote: Same as POQ

Feature Parameters for Order: Same as POQ

11.10.9 Monitoring

Definition: The Seller is monitoring the Internet Access for the Buyer.

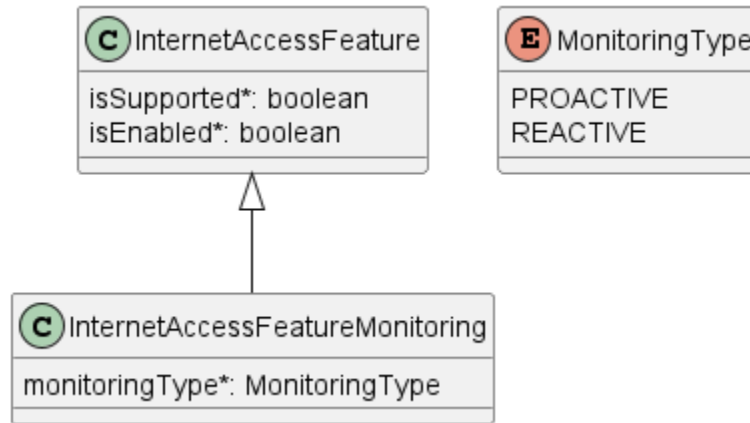


Figure 14 Monitoring Feature

Feature Parameters for POQ:

- *monitoringType*

Feature Parameters for Quote: Same as POQ

Feature Parameters for Order: Same as POQ

11.10.10 Performance Reporting

Definition: Performance Metrics are delivered to the Buyer for the Internet Access Product.

Feature Parameters for POQ: No additional parameters

Feature Parameters for Quote: No additional parameters

Feature Parameters for Order: No additional parameters

11.10.11 Routing

Definition: The method by which IP routes are exchanged between the Buyer and the Seller.

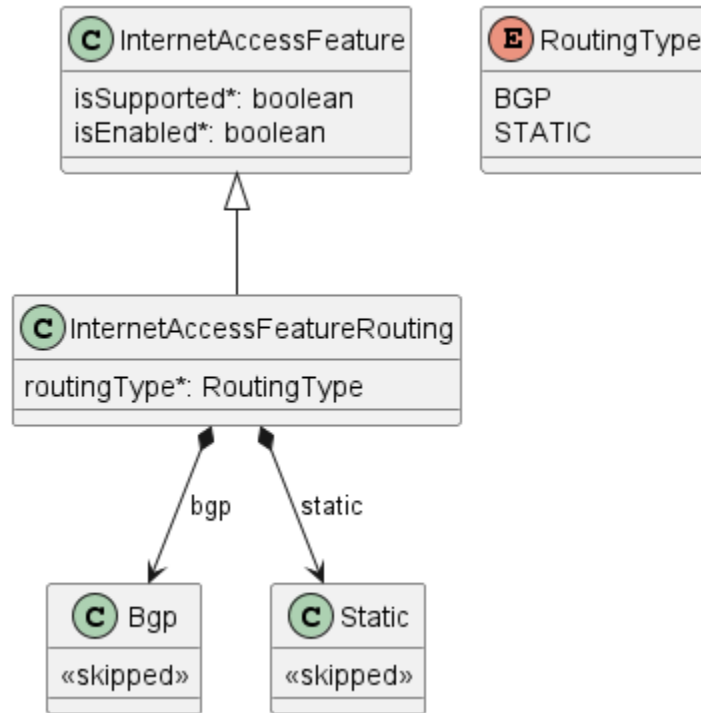


Figure 15 Routing Feature

Feature Parameters for POQ:

- *routingType*

Feature Parameters for Quote: Same as POQ

Feature Parameters for Order:

- *routingType*
- *static* – if *routingType*=*STATIC*, otherwise not provided
- *bgp* – if *routingType*=*BGP*, otherwise not provided

Editor Note 2: The model for BGP and Static routing protocols configuration is subject to change. This version contains a full model as defined in MEF 61.1 [7], which is going to be reviewed and potentially simplified for the context of Internet Access Product requirements.

11.10.11.1 Bgp

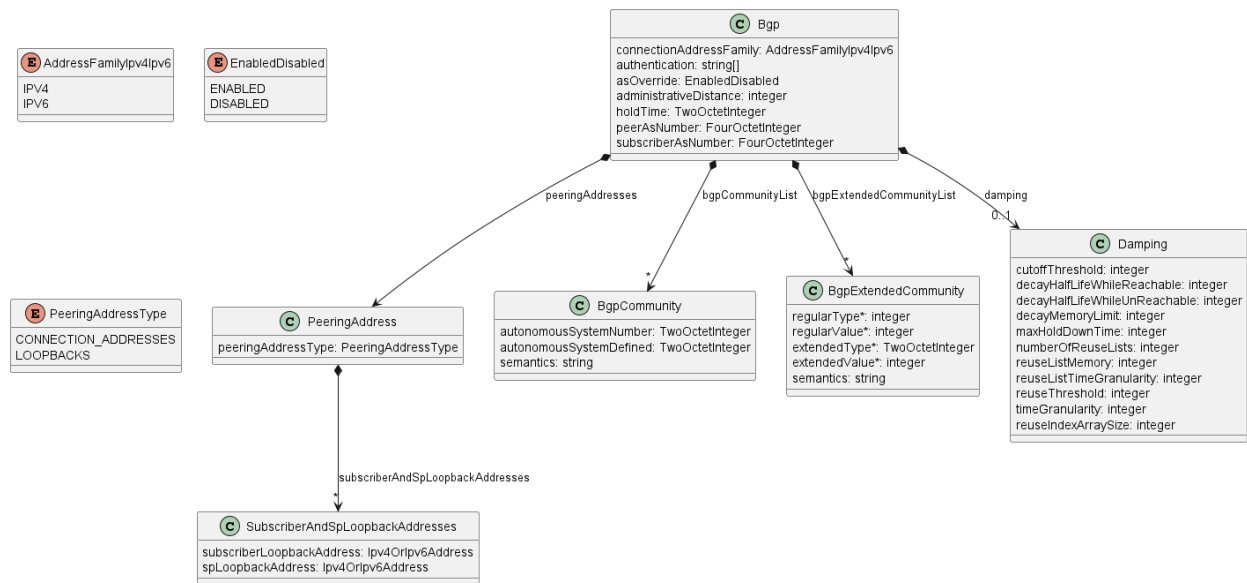


Figure 16 Bgp

Figure 16 depicts the model of BGP routing protocol configuration model.

File: schema\common\ip\ipRoutingProtocolsCommon.yaml



Internet Access Product Schema and Developer Guide

Name	Type	M/O	Description
subscriberAsNumber	FourOctetInteger	O	BGP Subscriber Autonomous System number.
peerAsNumber	FourOctetInteger	O	BGP Peer Autonomous System Number.
connectionAddressFamily	AddressFamilyIpv4Ipv6	O	Connection Address Family (IPv4 or IPv6).
peeringAddresses	PeeringAddress	O	Peering Addresses.
authentication	string[] maxItems=1	O	BGP Authentication. It is either empty or if present is it a value of MD5 Password. It is assumed that an encrypted channel is used when this data is passed across the API so that the password is protected.
bgpCommunityList	BgpCommunity[]	O	Used to control which routers are accepted, preferred, distributed, or advertised.
bgpExtendedCommunityList	BgpExtendedCommunity[]	O	Mechanism for labeling information carried in BGP-4. Provide enhancement over existing BGP Community Attribute an extended range, the addition of type field.
holdTime	TwoOctetInteger	O	Hold time in seconds. Indicates the agreed Hold Time used for BGP sessions. The possible values are 0 or an integer in the range 3-65535.
damping	Damping[] maxItems=1	O	Route flap damping. When the Damping parameter is empty, the attribute is not set. When not empty a single set of parameters described in Section 4.3 of RFC 2430 MUST be agreed.
asOverride	EnabledDisabled	O	Autonomous System Override. The SP (or Operator) can overwrite instances of the Subscriber's AS Number in the AS Path with their own AS Number, when advertising routes to the Subscriber. This needs to be explicitly agreed between the SP and the Subscriber, and/or between an SP/SO and an Operator.
administrativeDistance	integer minimum = 1	O	BGP Administrative Distance.

Table 12 Bgp

Example:

```

"routing": {
  "isSupported": true,
  "isEnabled": true,
  "routingType": "BGP",
  "bgp": {
    "subscriberAsNumber": 64512,
    "peerAsNumber": 65530,
    "connectionAddressFamily": "IPV4",
    "peeringAddresses": {
      "peeringAddressType": "CONNECTION_ADDRESSES"
    },
    "authentication": [],
    "bgpCommunityList": [],

```

```

"bgpExtendedCommunityList": [],
"holdTime": 180,
"damping": [
  {
    "cutoffThreshold": 10,
    "decayHalfLifeWhileReachable": 900,
    "decayHalfLifeWhileUnReachable": 1800,
    "decayMemoryLimit": 3600,
    "maxHoldDownTime": 3600,
    "numberOfReuseLists": 4,
    "reuseListMemory": 3600,
    "reuseListTimeGranularity": 300,
    "reuseThreshold": 2,
    "timeGranularity": 60,
    "reuseIndexArraySize": 10
  }
],
"asOverride": "DISABLED",
"administrativeDistance": 20
}
}

```

11.10.11.2 BgpCommunity

File: schema\common\ip\ipRoutingProtocolsCommon.yaml

A community is a group of destinations which share some common property. Each autonomous system administrator may define which communities a destination belongs to.

Name	Type	M/O	Description
autonomousSystemNumber	TwoOctetInteger	O	The first two octets encoding the Autonomous System value.
autonomousSystemDefined	TwoOctetInteger	O	The remaining octets.
semantics	string	O	Text describing how the Seller will handle routes tagged with this Community

Table 13 BgpCommunity

11.10.11.3 BgpExtendedCommunity

File: schema\common\ip\ipRoutingProtocolsCommon.yaml

This attribute provides a mechanism for labeling information carried in BGP-4. These labels can be used to control the distribution of this information, or for other applications.



Name	Type	M/O	Description
regularType	integer minimum = 0 maximum = 255	O	Regular Type Field, 1 octet length
regularValue	integer minimum = 0 maximum = 72057594037927935	O	Octets 2 - 8 of the value part of the address. Used in case only Regular Type is provided.
extendedType	TwoOctetInteger	O	Extended Type Field, 2 octets length
extendedValue	integer minimum = 0 maximum: 281474976710655	O	Octets 3 - 8 of the value part of the address. Used in case only Extended Type is provided.
semantics	string	O	Text describing how the Seller will handle routes tagged with this Community

Table 14 BgpExtentionCommunity

oneOf:

- required: [regularType, regularValue]

- required: [extendedType, extendedValue]

11.10.11.4 AddressFamilyIpv4Ipv6

File: schema\common\ip\ipRoutingProtocolsCommon.yaml

Specifies whether the session is established over IPv4 or IPv6.

Value
IPV4
IPV6

Table 15 AddressFamilyIpv4Ipv6

11.10.11.5 Damping

File: schema\common\ip\ipRoutingProtocolsCommon.yaml

Damping parameters as defined in RFC 2439 BGP Route Flap Damping, Section 4.2



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Name	Type	M/O	Description
cutoffThreshold	integer minimum = 0	O	This value is expressed as a number of route withdrawals. It is the value above which a route advertisement will be suppressed.
decayHalfLifeWhileReachable	integer minimum = 0	O	This value is the time duration in seconds during which the accumulated stability figure of merit will be reduced by half if the route is considered reachable (whether suppressed or not).
decayHalfLifeWhileUnReachable	integer minimum = 0	O	This value is the time duration in seconds during which the accumulated stability figure of merit will be reduced by half if the route is considered unreachable. If not specified or set to zero, no decay will occur while a route remains unreachable.
decayMemoryLimit	integer minimum = 0	O	This is the maximum time (in seconds) that any memory of previous instability will be retained given that the route's state remains unchanged, whether reachable or unreachable. This parameter is generally used to determine array sizes.
maxHoldDownTime	integer minimum = 0	O	This value is the maximum time a route can be suppressed no matter how unstable it has been prior to this period of stability. In seconds.
numberOfReuseLists	integer minimum = 0	O	This is the number of reuse lists. It may be determined from reuse-list-max or set explicitly.
reuseListMemory	integer minimum = 0	O	This is the time (in seconds) value corresponding to the last reuse list. This may be the maximum value of T-hold for all parameter sets of may be configured.
reuseListTimeGranularity	integer minimum = 0	O	This is the time (in seconds) interval between evaluations of the reuse lists. Each reuse lists corresponds to an additional time increment.
reuseThreshold	integer minimum = 0	O	This value is expressed as a number of route withdrawals. It is the value below which a suppressed route will now be used again.
timeGranularity	integer minimum = 0	O	This is the time granularity in seconds used to perform all decay computations.
reuseIndexArraySize	integer minimum = 0	O	This is the size of reuse index arrays. This size determines the accuracy with which suppressed routes can be placed within the set of reuse lists when suppressed for a long time.

Table 16 Damping

11.10.11.6 PeeringAddress

File: schema\common\ip\ipRoutingProtocolsCommon.yaml

Peering Addresses. Connection Addresses, or Loopbacks plus a list of pairs of IP addresses.
Reference - MEF 61.1 Section 12.7.3.



Name	Type	M/O	Description
peeringAddressType	PeeringAddressType	O	If the Peering Addresses parameter is CONNECTION_ADDRESSES, a separate BGP peering session is established over each UNI Access Link, using the primary IPv4 addresses in the UNI Access Link IPv4 Connection Addressing Service Attribute (section 13.4) or the first IPv6 addresses in the UNI Access Link IPv6 Connection Addressing Service Attribute (section 13.5), as indicated by the Connection Address Family parameter. If the Peering Addresses parameter is LOOPBACKS, a list of pairs of IP addresses is additionally specified, each pair containing the Subscriber's loopback address and the SP's or Operator's loopback address. A single BGP peering session is established for each pair of addresses.
subscriberAndSpLoopbackAddresses	SubscriberAndSpLoopbackAddresses[]	O	A list of pairs of IP addresses, each pair containing the Subscriber's loopback address and the SP's or Operator's loopback address. A single BGP peering session is established for each pair of addresses.

Table 17 PeeringAddress

11.10.11.7 PeeringAddressType

File: schema\common\ip\ipRoutingProtocolsCommon.yaml

If the Peering Addresses parameter is CONNECTION_ADDRESSES, a separate BGP peering session is established over each UNI Access Link, using the primary IPv4 addresses in the UNI Access Link IPv4 Connection Addressing Service Attribute (section 13.4) or the first IPv6 addresses in the UNI Access Link IPv6 Connection Addressing Service Attribute (section 13.5), as indicated by the Connection Address Family parameter. If the Peering Addresses parameter is LOOPBACKS, a list of pairs of IP addresses is additionally specified, each pair containing the Subscriber's loopback address and the SP's or Operator's loopback address. A single BGP peering session is established for each pair of addresses.

Value
CONNECTION_ADDRESSES
LOOPBACKS

Table 18 PeeringAddressType

11.10.11.8 SubscriberAndSpLoopbackAddresses

File: schema\common\ip\ipRoutingProtocolsCommon.yaml

A list of pairs of IP addresses, each pair containing the Subscriber's loopback address and the SP's or Operator's loopback address. A single BGP peering session is established for each pair of addresses.

Name	Type	M/O	Description
subscriberLoopbackAddress	Ipv4OrIpv6Address	O	Subscriber's loopback Address for BGP establishing a session
spLoopbackAddress	Ipv4OrIpv6Address	O	Service Provider's loopback Address for BGP establishing a session

Table 19 SubscriberAndSpLoopbackAddresses

11.10.11.9 Static

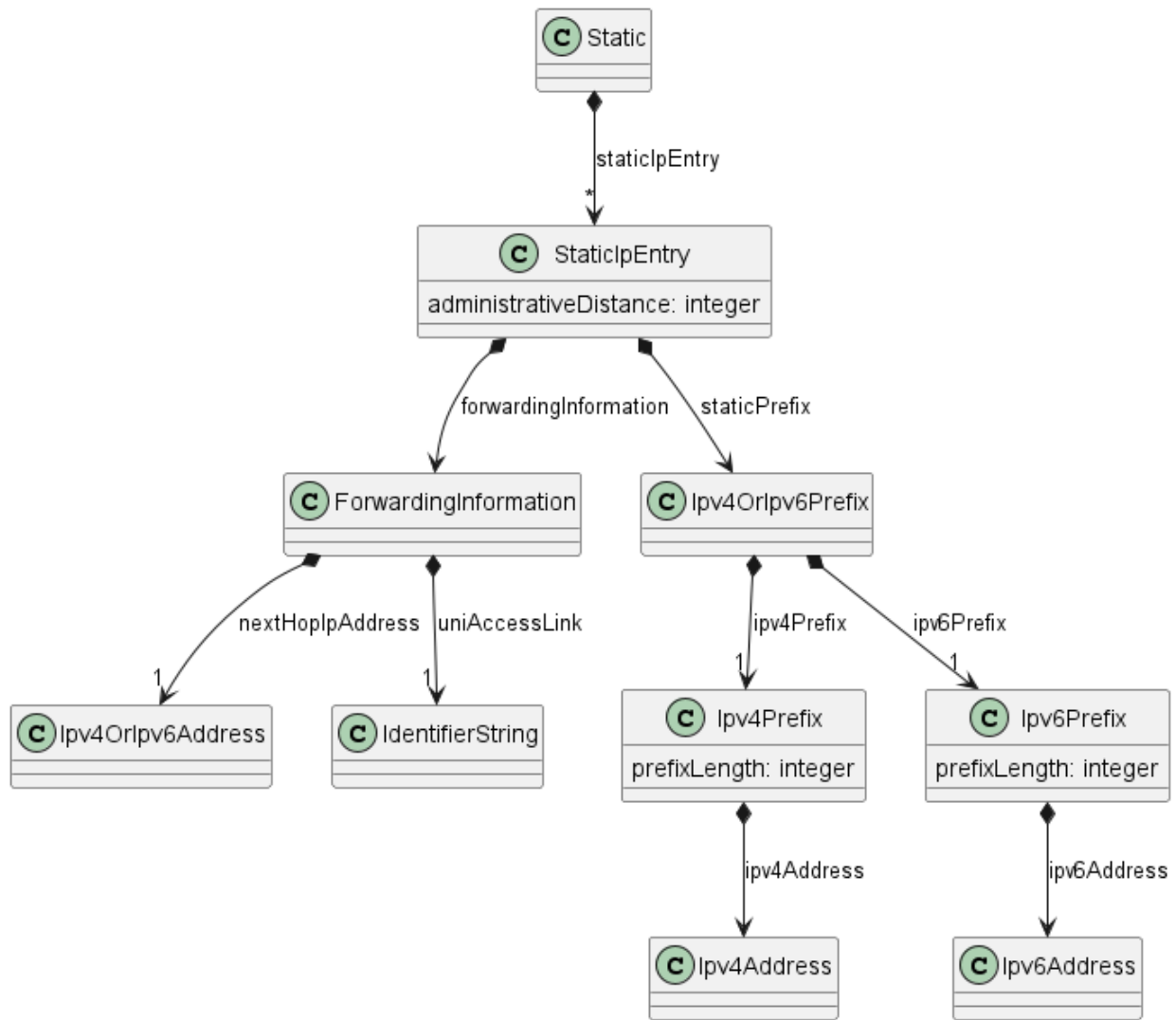


Figure 17 Static

Figure 17 shows the resource model for Static routing configuration.

File: schema\common\ip\ipRoutingProtocolsCommon.yaml

Name	Type	M/O	Description
staticIpEntry	StaticIpEntry[] minItems = 1	O	Static IP address entry.

Table 20 Static

Example:

```

"routing": {
  "isSupported": true,
  "isEnabled": true,
  "routingType": "STATIC",
  "static": {
    "staticIpEntry": [
      {
        "staticPrefix": {
          "ipv4Prefix": {
            "ipv4Address": "192.0.2.0",
            "prefixLength": 24
          }
        },
        "forwardingInformation": {
          "nextHopIpAddress": "198.51.100.1"
        },
        "administrativeDistance": 1
      },
      {
        "staticPrefix": {
          "ipv6Prefix": {
            "ipv6Address": "2001:db8::",
            "prefixLength": 48
          }
        },
        "forwardingInformation": {
          "nextHopIpAddress": "2001:db8::1"
        },
        "administrativeDistance": 1
      }
    ]
  }
}

```

11.10.11.10 StaticIpEntry

File: schema\common\ipRoutingProtocolsCommon.yaml

StaticIpEntry data type including IPv4/IPv6 prefixes, forwarding information and administrative distance.



Name	Type	M/O	Description
administrativeDistance	integer minimum = 1	O	Administrative distance, an integer > 0.
forwardingInformation	ForwardingInformation	O	Forwarding information with either Next Hop IP address or UNI Access Link identifier.
staticPrefix	Ipv4OrIpv6Prefix	O	IPv4 or IPv6 Prefix that is advertised.

Table 21 StaticIpEntry

11.10.11.11 ForwardingInformation

File: schema\common\ip\ipRoutingProtocolsCommon.yaml

Forwarding information, consisting of either a nexthop IP address in the Subscriber Network (if the access medium is multipoint capable, e.g., Ethernet), or a specific UNI Access Link (if the access medium is strictly point-to-point, e.g., HDLC, PPP over DSL).

Name	Type	M/O	Description
nextHopIpAddress	Ipv4OrIpv6Address	O	Next hop IP address.
uniAccessLink	IdentifierString	O	UNI Access Link identifier as set by the Buyer in IpUniAccessLink.identifier attribute.

Table 22 ForwardingInformation

oneOf:

- required: [nextHopIpAddress]

- required: [uniAccessLink]

11.10.12 SNMP Read

Definition: The Seller provides SNMP v2/v3 read access to the router including MIB-2.

Feature Parameters for POQ: No additional parameters

Feature Parameters for Quote: No additional parameters

Feature Parameters for Order: No additional parameters

11.11 Bandwidth

Applicable Use Cases: All

For Broadband Internet Access this Product Attribute represents the “nominal” bandwidth. For Dedicated Internet Access this Product Attribute represents the maximum bandwidth that can be sustained by the Buyer. The value of this Product Attribute is provided with the *BiDirSimpleBandwidth* type.

The term “nominal” is used to represent the name or description of the Internet Access Product. It is not necessarily a committed/guaranteed bandwidth or a maximum bandwidth although it is indicative of the expected or average bandwidth. For example, “a 400Mbps Internet Access”.

400Mbps is the nominal bandwidth and indicates that the Subscriber will be able to achieve approximately 400Mbps of bandwidth with some level of regularity.

File: schema\common\common.yaml

Type: BiDirSimpleBandwidth

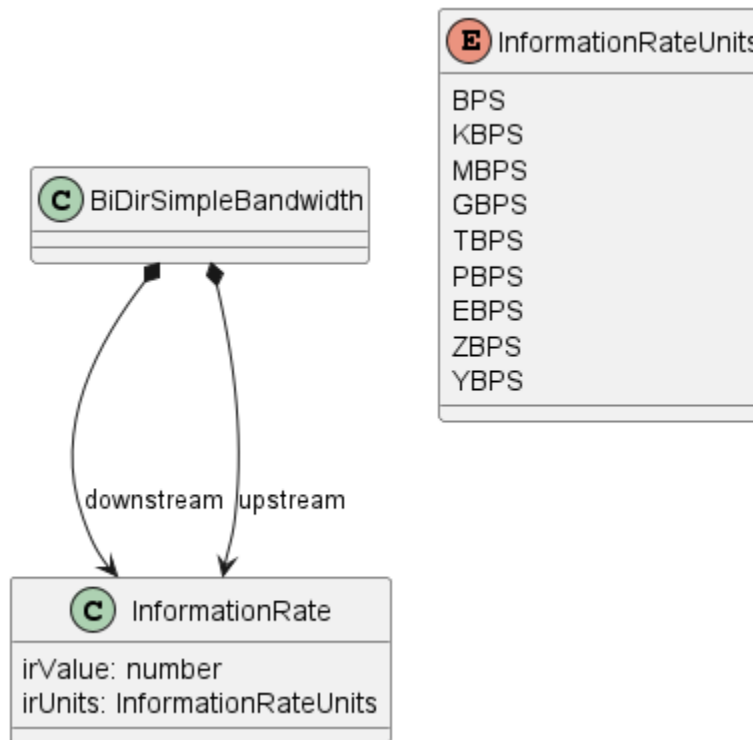


Figure 18 BiDirSimpleBandwidth

11.12 Guaranteed Bandwidth

Applicable Use Cases: All

The value of this Product Attribute is the bandwidth level that can be sustained by the Buyer on a continual basis with a very high probability of successful packet delivery. This parameter is commonly used for Dedicated Internet Access and rarely for Broadband Internet Access. The value of this Product Attribute is provided with the *BiDirSimpleBandwidth* type (see Figure 18)

[R19] The value of *guaranteedBandwidth* for each direction **MUST** be less than or equal to the value of *bandwidth* for the corresponding direction.

[O4] The *guaranteedBandwidth* **MAY NOT** be specified for Basic and Advanced Access Products.

11.13 Billing Model

Applicable Use Cases: All

The value of this Product Attribute indicates how the service is billed. Although the charge for the Product depends on the values of many Product Attributes, the billing model is based primarily on the bandwidth-related Product Attributes described in sections 11.11 and 11.12. Three models are defined.

Editor Note 3: Billing model overlaps and enhances the standard Pricing model covered by Quote and Order APIs. This modelling is subject to further study and may be covered by the Envelope APIs functions.

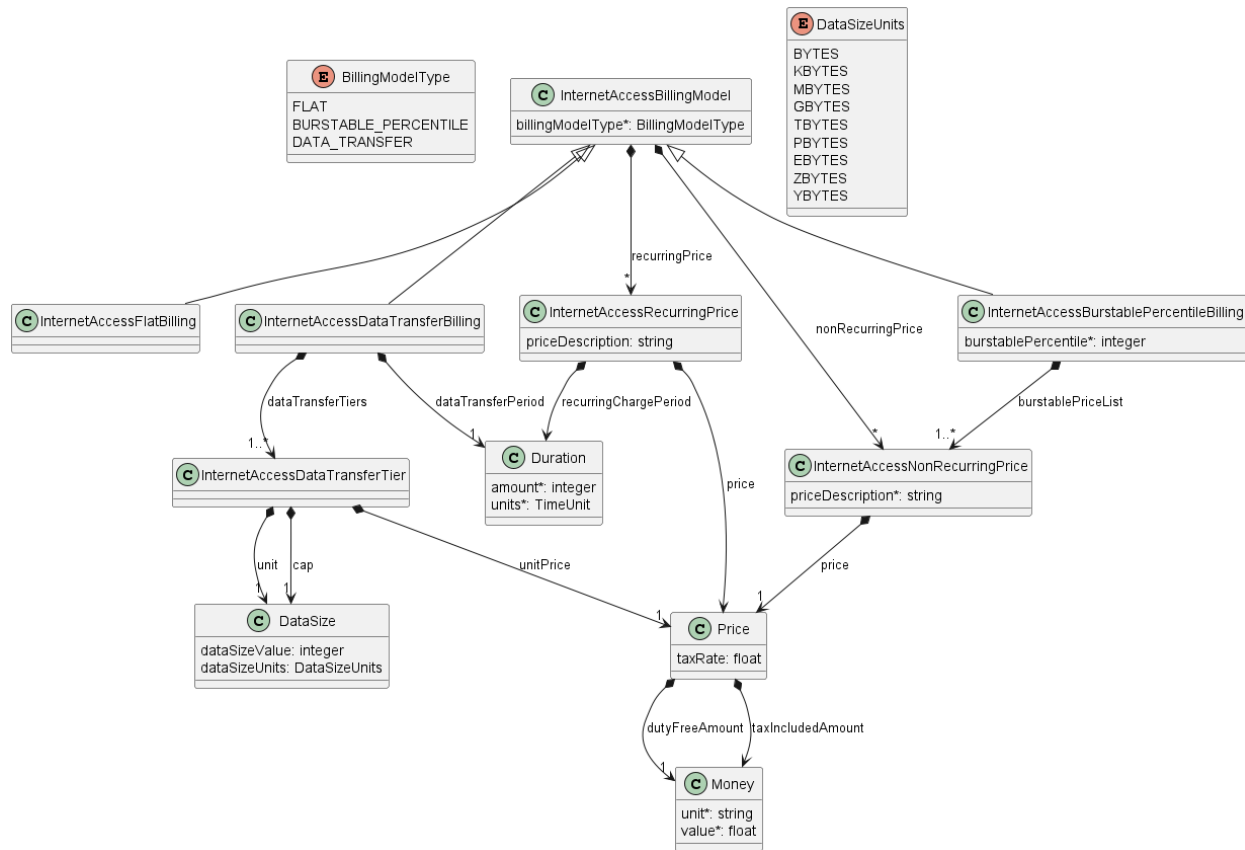


Figure 19 Billing Model

The *billingModel* values are defined as follows:

InternetAccessFlatBilling billing is a recurring amount for a fixed time period for a nominal amount of bandwidth. For example, \$30.00 per month for broadband Internet Access at 400 Mbps is a *InternetAccessFlatBilling* billing amount. For Dedicated Internet Access, this would be the maximum bandwidth available to the Buyer.

InternetAccessBurstablePercentileBilling billing is a recurring amount for a committed amount of bandwidth, the Burstable Bandwidth. For example, €20.00 per month for 400 Mbps of committed bandwidth and an additional amount per month for burstable bandwidth above that amount. The *burstablePercentile* specifies how this value is determined. A common value for this is 95 (95th Percentile).

InternetAccessDataTransferBilling is based on the amount of data rather than the rate. For example, during each *dataTransferPeriod*, this cost is \$10 per 10GB up to 100GB, and \$20.00 per 10GB for the next 500GB, and \$40 per 1GB for all additional data transfer. The value of the *dataTransferTiers* parameter specifies a list of values that describe each of these tiers. In the example above, there would be three entries, {10GB, 100GB}, the second would be {10GB, 500GB}, and the third would be {1GB, 0GB}.

[R20] The Internet Access Billing Model Product Attribute **MUST** be specified for Quote and Order.

Examples:

Flat Billing:

```
{
  "billingModelType": "FLAT",
  "nonRecurringPrice": [
    {
      "priceDescription": "Activation Fee",
      "price": {
        "dutyFreeAmount": { "unit": "USD", "value": 100.0 },
        "taxIncludedAmount": { "unit": "USD", "value": 123.0 },
        "taxRate": 23.0
      }
    }
  ],
  "recurringPrice": [
    {
      "priceDescription": "Monthly Access",
      "price": {
        "dutyFreeAmount": { "unit": "USD", "value": 30.0 },
        "taxIncludedAmount": { "unit": "USD", "value": 36.9 },
        "taxRate": 23.0
      },
      "recurringChargePeriod": { "amount": 1, "units": "months" }
    }
  ]
}
```

Burstable Billing:

```
{
  "billingModelType": "BURSTABLE_PERCENTILE",
  "nonRecurringPrice": [
    {
      "priceDescription": "Setup Fee",
      "price": {
        "dutyFreeAmount": { "unit": "EUR", "value": 200.0 },
        "taxIncludedAmount": { "unit": "EUR", "value": 246.0 },
```

```
1024         "taxRate": 23.0
1025     }
1026 }
1027 ],
1028 "recurringPrice": [
1029     {
1030         "priceDescription": "Committed Bandwidth",
1031         "price": {
1032             "dutyFreeAmount": { "unit": "EUR", "value": 20.0 },
1033             "taxIncludedAmount": { "unit": "EUR", "value": 24.6 },
1034             "taxRate": 23.0
1035         },
1036         "recurringChargePeriod": { "amount": 1, "units": "months" }
1037     }
1038 ],
1039 "burstablePercentile": 95,
1040 "burstablePriceList": [
1041     {
1042         "priceDescription": "Burstable Bandwidth",
1043         "price": {
1044             "dutyFreeAmount": { "unit": "EUR", "value": 5.0 },
1045             "taxIncludedAmount": { "unit": "EUR", "value": 6.15 },
1046             "taxRate": 23.0
1047         }
1048     }
1049 ]
1050 }
1051
1052 Data Transfer Billing:
1053 {
1054     "billingModelType": "DATA_TRANSFER",
1055     "nonRecurringPrice": [],
1056     "recurringPrice": [
1057         {
1058             "priceDescription": "Base Fee",
1059             "price": {
1060                 "dutyFreeAmount": { "unit": "USD", "value": 10.0 },
1061                 "taxIncludedAmount": { "unit": "USD", "value": 12.3 },
1062                 "taxRate": 23.0
1063             },
1064             "recurringChargePeriod": { "amount": 1, "units": "months" }
1065         }
1066     ],
1067     "dataTransferPeriod": { "amount": 1, "units": "months" },
1068     "dataTransferTiers": [
1069         {
```

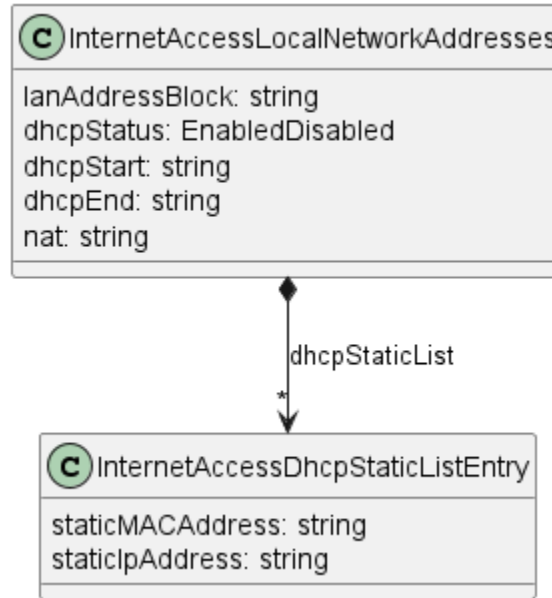
```
1070     "unit": { "dataSizeValue": 10, "dataSizeUnits": "GBYTES" },
1071     "cap": { "dataSizeValue": 100, "dataSizeUnits": "GBYTES" },
1072     "unitPrice": {
1073         "dutyFreeAmount": { "unit": "USD", "value": 10.0 },
1074         "taxIncludedAmount": { "unit": "USD", "value": 12.3 },
1075         "taxRate": 23.0
1076     }
1077 },
1078 {
1079     "unit": { "dataSizeValue": 10, "dataSizeUnits": "GBYTES" },
1080     "cap": { "dataSizeValue": 500, "dataSizeUnits": "GBYTES" },
1081     "unitPrice": {
1082         "dutyFreeAmount": { "unit": "USD", "value": 20.0 },
1083         "taxIncludedAmount": { "unit": "USD", "value": 24.6 },
1084         "taxRate": 23.0
1085     }
1086 },
1087 {
1088     "unit": { "dataSizeValue": 1, "dataSizeUnits": "GBYTES" },
1089     "cap": { "dataSizeValue": 0, "dataSizeUnits": "GBYTES" },
1090     "unitPrice": {
1091         "dutyFreeAmount": { "unit": "USD", "value": 40.0 },
1092         "taxIncludedAmount": { "unit": "USD", "value": 49.2 },
1093         "taxRate": 23.0
1094     }
1095 }
1096 ]
1097 }
```

11.14 Local Network Addresses

Applicable Use Cases:

- Broadband Internet Access, Advanced
- Dedicated Internet Access, with router

The Local Network Addresses attribute is used when the Seller is responsible for assigning Private Local Network Addresses for the Buyer's Network (see MEF 164 [18] section 11.3). The Product Attribute specifies the private address block to be assigned to the Buyer's IP hosts, whether DHCP is used, and if so, parameters necessary to configure the DHCP and NAT servers.


Figure 20 Local Network Addresses

Name	Type	M/O	Description
lanAddressBlock	string	O	IP address range, specified in CIDR notation.
dhcpStatus	EnabledDisabled	O	Indicates whether the Seller is providing a DHCP Service to centrally manage and automate the assignment of IP configurations for the Buyer's Network.
dhcpStart	string	O	Start of the DHCP address range. This is the first address in the range that can be assigned to a device. IPv4 or IPv6 address.
dhcpEnd	string	O	End of the DHCP address range. This is the last address in the range that can be assigned to a device. IPv4 or IPv6 address.
dhcpStaticList	InternetAccess-DhcpStaticListEntry	O	A list of static IP addresses that are reserved for specific devices in the network. Each entry in the list is a string representing an IPv4 or IPv6 address.
nat	string	O	An IP address if NAT is provided by the Seller

Table 23 Local Network Addresses

[R21] The *localNetworkAddresses* **MUST** be included in the Product Order if and only if the Seller is responsible for managing Local Network Addresses for the Buyer.



[R22] Values for *dhcpStart* and *dhcpEnd* **MUST NOT** be provided if and only if the value of *DHCP Status* is *Disabled*.

[R23] The value of *dhcpStaticList* **MUST** be an empty list if the value of *dhcpStatus* is *DISABLED*.

Requirements [R22] and [R23] ensure that values for DHCP parameters are specified only if the Seller is providing DHCP.

[R24] If the value of *dhcpStatus* is *ENABLED*, the IP addresses specified as values of *dhcpStart* and *dhcpEnd*, and any IP address specified in the *staticIpAddress* element of the *dhcpStaticList*, **MUST** be in the block of IP Addresses specified by *lanAddressBlock*.

[R25] If the value of *dhcpStatus* is *ENABLED*, the IP address specified in the value of *dhcpEnd* **MUST** be numerically greater than the IP address specified in the value of *dhcpStart*.

[R26] The value of *nat*, if specified, **MUST** be one of the Public Addresses used to access the Buyer's network.

Example:

```
"localNetworkAddresses": {  
  "lanAddressBlock": "10.0.0.0/24",  
  "dhcpStatus": "ENABLED",  
  "dhcpStart": "10.0.0.10",  
  "dhcpEnd": "10.0.0.200",  
  "dhcpStaticList": [  
    {  
      "staticMACAddress": "00:11:22:33:44:55",  
      "staticIpAddress": "10.0.0.50"  
    },  
    {  
      "staticMACAddress": "00:11:22:33:44:66",  
      "staticIpAddress": "10.0.0.51"  
    }  
  ],  
  "nat": "203.0.113.5"  
}
```

11.15 Provider Independent IP Addresses

Applicable Use Cases:

- Broadband Internet Access, Advanced
- Dedicated Internet Access with and without router

This Product Attribute indicates whether the Buyer is using Public (WAN) IP addresses routable over the Internet purchased directly from a registration authority for Internet Access or Public IP addresses obtained from the Seller.

Type: EnabledDisabled

ENABLED, meaning that the Buyer is providing the addresses

DISABLED, meaning that the Seller is providing the addresses

11.16 Autonomous System Number

Applicable Use Cases:

- Broadband Internet Access, Advanced
- Dedicated Internet Access, with and without router

An autonomous system (AS) is a collection of connected Internet Protocol (IP) routing prefixes under the control of one or more network operators on behalf of a single administrative entity or domain, that presents a common and clearly defined routing policy to the Internet. If the Buyer is providing the Internet-routable IP Addresses (see 11.15) then this Product Attribute is required to be specified. The value of this Product Attribute has two components, the type of Autonomous System and the Autonomous System Number (ASN).

Name	Type	M/O	Description
asnType	AsnType	O	The type of the ASN. This MUST be provided If the value of the 'providerIndependentIpAddress' is Enabled
asnId	FourOctetInteger	O	ASN Id. This MUST be provided if the value of asnType is 'PUBLIC'

Table 24 Autonomous System Number

[D2] If the value of the *providerIndependentIpAddresses* is *ENABLED*, the *autonomousSystemNumber* **SHOULD** be specified.

[R27] If the value of the *asnType* is *PUBLIC*, then the *asnId* **MUST** be provided by the Buyer.

11.17 IP Address Type

Applicable Use Cases:

- Broadband Internet Access, Advanced
- Dedicated Internet Access, with and without router

The value of this Product Attribute indicates whether the publicly routable IP address block consists of fixed addresses or addresses that can change over time.

Value
STATIC
DYNAMIC

Table 25 InternetAccessIpAddressType

[R28] The *ipAddressType* **MUST** be specified for Quote and Order.

[R29] If the value of the *providerIndependentIpAddresses* is *ENABLED*, the value of the *ipAddressType* **MUST** be *STATIC*.

11.18 Public Addresses Prefix Length

Applicable Use Cases:

- Broadband Internet Access, Advanced
- Dedicated Internet Access, with and without router

If the Seller is providing the Public Addresses for the Internet Access Product (i.e., the value of the *providerIndependentIpAddresses* is *DISABLED*) then this Product Attribute specifies the size of the IPv4 address block and the IPv6 address block requested by the Buyer and provided by the Seller. The value of this Product Attribute has two components: a string that indicates the length (in bits) of the network mask for the IPv4 address block and a string that indicates the length of the network mask for the IPv6 address block. By extension, these strings indicate the length of the host mask for each address block.

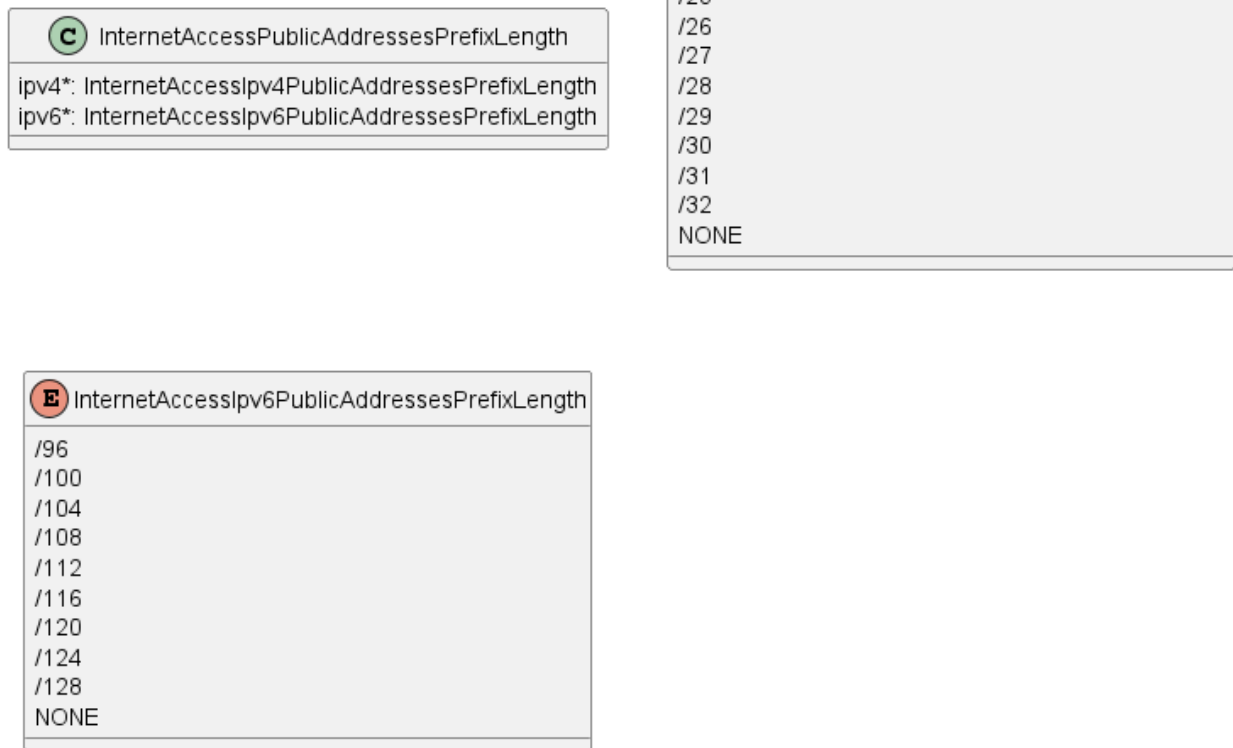


Figure 21 InternetAccessPublicAddressesPrefixLength

The values “/32” and “/128”, respectively, indicate a host route, i.e., a single IP address. This is common for Basic Internet Access and Advanced Internet Access. The value *NONE* indicates that no addresses should be allocated for the specified IP version.

[R30] If the value of the *providerIndependentIpAddresses* is *ENABLED*, the *publicAddressesPrefixLength* **MUST NOT** be provided.

[R31] If the value of the *providerIndependentIpAddresses* is *DISABLED*, *publicAddressesPrefixLength* **MUST** be provided for Quote and Order.

[R32] If the value of the *providerIndependentIpAddresses* is *DISABLED*, at least one of the components of *providerIndependentIpAddresses* (*ipv4* or *ipv6*) **MUST NOT** have the value *NONE*.

11.19 Public Address Block

Applicable Use Cases:

- Broadband Internet Access, Advanced
- Dedicated Internet Access, with and without router

If the Buyer is providing the WAN (public) addresses for the Internet Service (i.e., the value of the *providerIndependentIpAddresses* is *ENABLED*), then this Product Attribute fully specifies the blocks (IPv4 or IPv6) of public addresses used by the Buyer, e.g., “72.23.28.248/29”.

Name	Type	M/O	Description
ipv4*	string <i>nullable = true</i>	O	IPv4 Address Block in CIDR notation. `null` value represents the value of `None`
ipv6*	string <i>nullable = true</i>	O	IPv6 Address Block in CIDR notation. `null` value represents the value of `None`

Table 26 InternetAccessPublicAddressBlock

[R33] If the value of the *providerIndependentIpAddresses* is *DISABLED*, the *publicAddressBlock* **MUST NOT** be provided.

[R34] If the value of the *providerIndependentIpAddresses* is *ENABLED*, the *publicAddressBlock* **MUST** be specified for Quote and Order.

[R35] If the value of *providerIndependentIpAddresses* is *ENABLED*, at least one of the components of the *publicAddressesPrefixLength* (*ipv4* or *ipv6*) **MUST NOT** have the value *null*.

11.20 Transit Addresses

Applicable Use Cases:

- Dedicated Internet Access, with and without router

Internet Access Transit Addresses Prefix Length Product Attribute specifies the size of the IPv4 address block and the IPv6 address block used for Transit Addresses (see MEF 164 [18] section 11.2) in the Seller’s network to access the Buyer’s Public Addresses. The value of this Product Attribute allows the Buyer to specify the number of Transit Addresses (IPv4 and IPv6) that it wants the Seller to allocate and the Seller to return the Transit Address blocks allocated by the Seller to the Buyer.

InternetAccessTransitAddresses
ipv4TransitAddressPrefixLength: InternetAccessIpv4TransitAddressPrefixLength ipv6TransitAddressPrefixLength: InternetAccessIpv6TransitAddressPrefixLength ipv4TransitAddressBlock: string ipv6TransitAddressBlock: string

InternetAccessIpv4TransitAddressPrefixLength
/24 /25 /26 /27 /28 /29 /30 /31 /32 NONE

InternetAccessIpv6TransitAddressPrefixLength
/120 /124 /128 NONE

Figure 22 InternetAccessTransitAddresses

Name	Type	M/O	Description
ipv4TransitAddress-PrefixLength	InternetAccessIpv4TransitAddress-PrefixLength	O	IPv4 Transit Address Prefix Length
ipv6TransitAddress-PrefixLength	InternetAccessIpv6TransitAddress-PrefixLength	O	IPv6 Transit Address Prefix Length
ipv4TransitAddress-Block	string	O	IPv4 address block, specified in CIDR notation or NONE
ipv6TransitAddress-Block	string	O	IPv6 address block, specified in CIDR notation or NONE

Table 27 InternetAccessTransitAddresses

A common case is that the Buyer requests a single IPv4 Transit Address (by specifying “/32”) and the Seller responds with a single IPv4 address (e.g., 71.71.32.32/32)

Example:

```

"transitAddresses": {
  "ipv4TransitAddressPrefixLength": "/30",
  "ipv6TransitAddressPrefixLength": "/124",
  "ipv4TransitAddressBlock": "198.51.100.8/30",
  "ipv6TransitAddressBlock": "2001:db8:0:1::/124"
}

```

12 References

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1273 **Appendix A Usage examples (Informative)**

1274 *Editor Note 4: Once the proposed model goes through the Call for Comments lifecycle, an*
1275 *appendix with comprehensive examples throughout all APIs will be provided*
1276 *here. This version provides only standalone payload examples.*